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AMPK-endoplasmic reticulum stress axis contributes to lipopolysaccharide-caused mitochondrial dysfunction by regulating mitochondria-associated membrane function in bovine hepatocytes

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ABSTRACT

Mitochondrial homeostasis is closely associated with cellular homeostasis process, whereas mitochondrial dysfunction contributes to apoptosis and mitophagy. Hence, analyzing the mechanism of lipopolysaccharide (LPS)-caused mitochondrial damage is necessary to understand how cellular homeostasis is maintained in bovine hepatocytes. Mitochondria-associated membranes (MAM), a connection between endoplasmic reticulum (ER) and mitochondria, is important to control mitochondrial function. To investigate the underlying mechanisms of the LPS-caused mitochondrial dysfunction, hepatocytes isolated from dairy cows at ~160 d in milk (DIM) were pretreated with the specific inhibitors of adenosine 5′-monophosphate-activated protein kinase (AMPK), ER stress, RNA-activated protein kinase-like ER kinase (PERK), inositol-requiring enzyme 1α (IRE1α), c-Jun N-terminal kinase, and autophagy followed by a 12 μg/mL LPS treatment. The results showed that inhibiting ER stress with 4-phenylbutyric acid decreased the levels of autophagy and mitochondrial damage with AMPK inactivation in LPS-treated hepatocytes. The AMPK inhibitor compound C pretreatment alleviated LPS-induced ER stress, autophagy and mitochondrial dysfunction by regulating the expression of MAM-related genes, such as mitofusin 2 (MFN2), PERK, and IRE1α. Moreover, inhibiting PERK and IRE1α mitigated autophagy and mitochondrial dynamic disruption by regulating the MAM function. Additionally, blocking c-Jun N-terminal kinase, the downstream sensor of IRE1α, could reduce the levels of autophagy and apoptosis and restore the balance of mitochondrial fusion and fission by modulating the B cell leukemia 2 (BCL-2)/BCL-2 interacting protein 1 (BECLIN1) complex in the LPS-treated bovine hepatocytes. Furthermore, autophagy blockage with chloroquine could intervene in LPS-caused apoptosis to restore mitochondrial function. Collectively, these findings suggest that the AMPK-ER stress axis is involved in the LPS-caused mitochondrial dysfunction by mediating the MAM activity in bovine hepatocytes.

Key words: mitochondrial dysfunction, adenosine 5′-monophosphate-activated protein kinase, endoplasmic reticulum stress, mitochondria-associated membranes, bovine hepatocytes

INTRODUCTION

The dominant endotoxin of gram-negative bacteria, LPS, can cause hepatic injury and inflammatory response in the liver of ruminants fed a high-concentrate diet (Chang et al., 2015; Xu et al., 2015, 2016). Hepatocytes are rich in mitochondria because of the central role of the liver in carbohydrate, lipid, and protein metabolic homeostasis (McLaughlin et al., 2020). Evidence suggests that mitochondria play a vital role in maintaining hepatic homeostasis, and mitochondria dysfunction is closely associated with liver diseases (Teckman et al., 2004). It has been well documented that mitochondria are the central executioners of cell fate, including apoptotic and necrotic cell death. Mitochondria are the highly dynamic organelles; hence, controlling mitochondrial quality becomes an important issue to avoid cell death (Ni et al., 2015). Aberrant mitochondrial fission results in mitochondrial depolarization, autophagy or mitophagy, or apoptosis (Twig et al., 2008; Twig and Shirihai, 2011; Wang et al., 2017). These publications indicate that mitochondrial function is important to maintain liver function, but the underlying molecular mechanisms of LPS-caused mitochondrial dysfunction in bovine hepatocytes need to be clarified.

Under metabolic stress, accumulation of the unfolded/misfolded proteins in the endoplasmic reticulum

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(**ER**) lumen results in ER stress. The accumulating unfolded or misfolded proteins bind to glucose-regulated protein 78 (**GRP78**), causing it to dissociate from RNA-activated protein kinase-like ER kinase (**PERK**), inositol-requiring enzyme 1α (**IRE1α**), and activating transcription factor 6 (**ATF6**), followed by the initiation of unfolded protein response (**UPR**) pathways to activate autophagy and apoptosis (Senft and Ronai, 2015). The close association between ER and mitochondria originates dynamic platforms, termed mitochondria-associated membranes (**MAM**). The MAM provide an effective scaffold for the efficient interorganelle exchange of Ca^{2+} ions and lipids, which is important to mediate mitochondrial bioenergetics and prosurvival/prodeath pathways (Giorgi et al., 2015). In response to cellular stress, autophagy, a lysosome-dependent cellular degradation pathway, is initiated for cell survival and homeostasis maintenance. However, overactivated autophagy also contributes to cell apoptosis and cell damage (Ravanan et al., 2017). Hence, exploring whether ER stress-mediated MAM function and autophagy contributes to LPS-caused mitochondrial damage in bovine hepatocytes might help to mitigate the negative effect of LPS on liver. Previous studies have demonstrated that adenosine monophosphate-activated protein kinase (**AMPK**) signaling pathway participates in cellular homeostasis by regulating ER stress, autophagy, and mitochondrial biogenetics (Sun et al., 2019; Fan et al., 2020; Hu et al., 2021). These studies demonstrate that AMPK is involved in cellular homeostasis maintenance by mediating ER stress, autophagy, and mitochondrial function. However, the role of AMPK in the LPS-induced disruption of bovine hepatocyte homeostasis is unknown.

Therefore, we hypothesize LPS treatment causes mitochondrial damage by regulating ER stress to disrupt cellular homeostasis, and AMPK plays a vital role in determining the effect of ER stress on LPS-caused mitochondrial dysfunction in bovine hepatocytes. Inhibiting AMPK and ER stress can alleviate the LPS-caused disruption of mitochondrial homeostasis. Our study provides a theoretical basis for novel insights into alleviating the negative effect of LPS on the liver of dairy cows.

MATERIALS AND METHODS

Reagent and Chemicals

Escherichia coli O111:B4 lyophilized powder was used as a source of LPS herein (L2630 from Sigma). The ER stress inhibitor 4-phenylbutyric acid (**4-PBA**), the AMPK inhibitor (compound C; **CC**), the PERK inhibitor (GSK2656157), the IRE1 inhibitor (4μ8C), the JNK inhibitor (SP600125), and the autophagy inhibitor (chloroquine, **CQ**) were all purchased from MedChemExpress.

Bovine Hepatocyte Isolation, Culture, and Identification

Primary bovine hepatocytes were kindly provided and shipped on dry ice by Juan J. Loor (Animal Science Department and Nutritional Science Division, University of Illinois, Urbana). The hepatocytes were frozen in liquid nitrogen as soon as we received them. The cell viability of thawed cells was estimated by trypan blue after cryopreservation, and it was found to be decreased by less than 7%. A nonperfusion technique was used to isolate bovine hepatocytes from the liver tissue (1–3 g), which was harvested from 3 mid-lactation multiparous Holstein cows (approximately 160 d postpartum, 31.25 ± 2.1 kg/d of milk production, 1.7 parity, and 162 ± 3 DIM) using puncture biopsy as described (Zhou et al., 2018). After immersion in 70% ethanol for 30 s, the liver tissue was transported to the laboratory and was being finely minced with a scalpel blade. The tissue was washed with Ca^{2+}- and Mg^{2+}-free Hanks' balanced salt solution (**HBSS**; Corning) and centrifuged at 50 × *g* for 3 min at room temperature. The minced tissue was then resuspended in 50 mL of HBSS containing 1 m*M* calcium chloride and 150 U/mL type I collagenase (Sigma Chemical Co.), followed by the incubation with collagenase I for 40 min at 37°C with gentle shaking. Subsequently, the collagenase digestion was stopped using 25 mL of HBSS containing 10% fetal bovine serum (Thermo Fisher Scientific). The cell suspension was filtered through a 200-mesh cell strainer and washed with PBS for 3 times before being resuspended in a growth medium comprising Dulbecco's modified eagle medium (**DMEM**), 10% fetal bovine serum, 1 n*M* glucagon, 10 n*M* dexamethasone, 10 ng/mL epidermal growth factor, 10 n*M* insulin, and 1% penicillin/streptomycin. Hepatocytes were plated in T75 flasks at 3×10^5 cells per flask. All medium supplements were purchased from HyClone. Cells were maintained in a 37°C humidified cell incubator containing 95% air and 5% CO_2. The expression of the hepatocyte-specific marker cytokeratin-18 (**CK18**) was used to identify the cells.

Experimental Design

Lipopolysaccharide (12 μg/mL) was used to treat the bovine hepatocytes for 12 h to induce ER stress, autophagy, and apoptosis as previously demonstrated (Xie et al., 2023). To investigate the effect of ER stress on LPS-induced autophagy and mitochondrial dysfunction, the hepatocytes were exposed to different doses

of the ER stress inhibitor 4-PBA for 12 h to assess the effect of 4-PBA on the cell viability. Then, 1 m*M* dose of 4-PBA was used to pretreat the hepatocytes for 12 h, followed by the LPS treatment for another 12 h. The following groups were included in the experiment: control (n = 3), LPS-treated (n = 3), 4-PBA-treated (n = 3), and 4-PBA-treated followed by LPS treatment (**PBA+LPS**; n = 3). The AMPK inhibitor CC was used to determine the molecular mechanisms of AMPK in regulating ER stress, autophagy, and mitochondrial dynamic disruption caused by LPS. The hepatocytes were exposed to different concentrations of CC for 9 h to check the influence of CC on the cell viability. Subsequently, the hepatocytes were exposed to 1 μ*M* CC for 9 h followed by the LPS treatment for another 12 h. The test was divided into 4 groups: control (n = 3), LPS-treated (n = 3), CC-treated (n = 3), and CC-treated followed by LPS treatment (**CC+LPS**; n = 3). To investigate the role of the PERK and IRE1/JNK pathways in LPS-caused autophagy and mitochondrial homeostasis disruption, the PERK inhibitor (GSK2656157), the IRE1 inhibitor (**4μ8C**), and the JNK inhibitor (SP600125) were chosen. Hepatocytes were treated with different doses of GSK2656157, 4μ8C and SP600125 for 12 h. Doses of inhibitors (1 μ*M* GSK2656157, 10 μ*M* 4μ8C, and 10 μ*M* SP600125) were selected to pretreat the hepatocytes for 12 h, followed by LPS exposure for another 12 h. For GSK2656157 treatment, the experiment was divided into 4 groups: control (n = 3), LPS-treated group (n = 3), GSK2656157-treated (**GSK**; n = 3), and GSK2656157-treated followed by LPS treatment (**GSK+LPS**; n = 3). For 4μ8C treatment, 4 groups were included: control (n = 3), LPS-treated (n = 3), 4μ8C-treated (n = 3), and 4μ8C-treated followed by LPS treatment (**4μ8C+LPS**; n = 3). For SP600125 treatment, the groups were contained 4 groups: control (n = 3), LPS-treated (n = 3), SP600125-treated (**SP125**; n = 3), and SP600125-treated followed by LPS treatment (**SP125 +LPS**; n = 3). To study the effect of autophagy on mitochondrial damage in the LPS-treated hepatocytes, the autophagy inhibitor CQ was used to pretreat hepatocytes followed by the LPS treatment for another 12 h. The hepatocytes were treated with different concentrations of CQ for 6 h to test the influence of CQ on the cell viability, and 1 μ*M* CQ was used to treat the hepatocytes. The experiment was divided into 4 groups: control (n = 3), LPS-treated (n = 3), CQ-treated (n = 3), and CQ-treated followed by LPS treatment (**CQ +LPS**; n = 3).

Hepatocytes were plated in 96-, 24-, and 6-well microplates at different densities for different assays as in the previous studies with minor modifications (Xu et al., 2018; Roy et al., 2019). Hepatocytes were seeded in each well of the 96-well microplate at the density of 1 $\times$ 10^4 for cell viability assay. A total of 5 $\times$ 10^4 bovine hepatocytes were grown in each well of the 24-well plate for quantitative reverse-transcription (**qRT**)-PCR assay, immunofluorescence analysis, Mito-tracker staining, Lyso-tracker staining, and JC-1 staining. Then 3 $\times$ 10^5 bovine hepatocytes were seeded in each well of the 6-well microplate and divided into different groups based on the experimental design. After the indicated treatment, the cells were harvested for western blot assay.

Cell Viability

A total of 1 $\times$ 10^4 bovine hepatocytes were seeded in each well of 96-well microplate and were grown in a 37°C humidified cell incubator with 95% air and 5% CO_2 for 24 h. Hepatocytes were exposed to different doses of 4-PBA for 12 h, CC for 9 h, GSK157 for 12 h, 4μ8C for 12 h, SP125 for 12 h, and CQ for 6 h. Each treatment was repeated for 6 times. After the indicated treatment, 10 μL of a cell counting kit-8 solution (Solarbio Life Sciences) was added into each well to measure the live cell percentage, and the microplates were incubated for 4 h at 37°C. The absorption was determined at 450 nm using a microplate photometer (Thermo Fisher Scientific Inc.).

Extraction of RNA and Performance of qRT-PCR

After the indicated treatment, the total RNA of bovine hepatocytes was extracted using the RNA iso plus reagent (Takara) according to the manufacture's protocol. Approximately 500 ng of RNA was mixed with HiScript III RT SuperMix (R323–01, Vazyme Biotech) to synthesize cDNA by reverse transcription as directed by the manufacture. Details of target primers are listed in Supplemental Table S1 (https://data.mendeley.com/datasets/f8rhxzkvvh/1; Xie, 2023). The qRT-PCR was performed using ChamQ Universal SYBR qPCR Master Mix (Q711, Vazyme) on the ABI 7300 system. Glyceraldehyde phosphate dehydrogenase (*GAPDH*) served as the internal reference gene for the normalization of gene expression. The validity of *GAPDH* for a housekeeping gene in the liver of cows has been reported in the previous study (Xie et al., 2022). The $2^{-\Delta\Delta Ct}$ method was used to analyze the relative expression of target genes.

Western Blot Analysis

After the indicated treatment, the total protein of hepatocytes was extracted using the radioimmunoprecipitation assay lysis buffer with 1 m*M* phenylmethlsulfonyl fluoride. The protein concentration was measured

using a commercial bicinchoninic protein assay kit. Subsequently, the samples were diluted to the same concentration and denatured with a 5 × SDS loading buffer at 99°C for 5 min. An equal amount of protein was separated using the 10% (12.5% for LC3 protein separation) SDS-PAGE by electrophoresis and transferred onto a polyvinylidene fluoride membrane. The membrane was blocked with 7% skim milk (5% BSA in case of phosphorylated proteins) for 2 h at room temperature. The membranes were then incubated with primary antibodies at 4°C overnight. After being washed with Tris-buffered saline containing Tween 20 (TBST) 3 times, the membranes were incubated with HRP-conjugated secondary antibody at room temperature for 2 h. Bands on the membrane were visualized using an Enhanced Chemiluminescence Plus kit (Vazyme), and the signals were recorded using a ChemiDoc MP system (Bio-Rad). Image Lab software (Bio-Rad) was used to analyze the intensity of each band. Glyceraldehyde phosphate dehydrogenase (GAPDH) or β-actin (ACTB) was used as the reference protein for the normalization of the target protein abundance. The primary and second antibody details are listed in the Supplemental Table S2 (https://data.mendeley.com/datasets/f8rhxzkvvh/1; Xie, 2023).

Immunofluorescence Analysis

Hepatocytes were plated on circular coverslips in the wells of the 24-well microplate. After the indicated treatment, the cells were fixed with 4% paraformaldehyde (Sigma-Aldrich) for 15 min at room temperature. After 3 gentle washes with PBS, the cells were incubated with 0.3% Triton X-100 (Solarbio) for 15 min at room temperature to improve permeability. Subsequently, 3% BSA served as the blocking agent to block the hepatocytes for 1 h at room temperature. Then, the coverslips were incubated with the primary antibodies at 4°C for 16 h. The cells were then incubated with the fluoresceine isothiocyanate (FITC)- or Cy3-labeled secondary antibodies for 1 h at 37°C in the dark. After 3 gentle washes with PBS containing Tween 20 (PBST), 4′,6-diamidino-2-phenylindole (DAPI) staining solution (Solarbio) was used to counter-stain nuclei in the dark for 10 min. The coverslips were mounted on glass slides using antifade mounting medium and visualized under a laser scanning microscope (LSM) 710 confocal laser microscope system (Zeiss). The primary and second antibody details are listed in the Supplemental Table S2.

Mito-tracker, Lyso-tracker, and JC-1 Staining

Bovine hepatocytes were grown on circular coverslips in the 24-well plates. After the indicated incubation, the cells were incubated with 100 n*M* Mito-tracker red CMXRos (40741ES50, Yeasen) for 15 min, a 50 n*M* Lyso-tracker red solution (C1046, Beyotime) for 30 min, a 10 μg/mL JC-1 staining solution (40705ES03, Yeasen) for 15 min at 37°C in the dark. After washing with PBS, the cells were observed under a laser scanning microscope (LSM) 710 confocal laser microscope system (Zeiss).

Statistical Analysis

All data were analyzed using one-way ANOVA with Dunnett's post hoc test by IBM SPSS 20.0 Statistics for Windows (IBM Inc.). Before comparing the differences between groups, the normality distribution of the variables was assessed using the Shapiro-Wilk test. Results are expressed as the mean and standard error of the mean (mean ± SEM). Differences were considered significant for $P < 0.05$ or highly significant for $P < 0.01$. All the graphs were created using the GraphPad Prism 8 software for Windows (GraphPad Software).

RESULTS

Identification of Bovine Hepatocytes

The expression of CK18 was detected to identify the bovine hepatocytes. As shown in Supplemental Figure S1 (https://data.mendeley.com/datasets/f8rhxzkvvh/1; Xie, 2023), CK18 showed a strong immunofluorescence intensity in the cells, indicating that the phenotype of the cells used in the current study was the same as that of the hepatocytes.

AMPK Is Involved in LPS-Induced ER Stress in Bovine Hepatocytes

As shown in Figure 1a, the hepatocytes were treated with 1∼10 m*M* 4-PBA for 12 h, and the cell viability was significantly reduced at the doses of 2, 4, 8, and 10 m*M* ($P < 0.01$). Hence, we chose the 1 m*M* dose 4-PBA to treat the cells. Compared with that in the LPS group, inhibiting ER stress with 4-PBA could downregulate the mRNA relative expression of ER stress-related genes, such as *GRP78*, *IRE1*, TNF receptor-associated factor 2 (*TRAF2*), c-Jun N-terminal kinase 1 (*JNK1*), *PERK*, C/EBP homologous protein (*CHOP*), and *ATF6* ($P < 0.05$, Figure 1b). Consistent with the mRNA results, the protein levels of GRP78, p-PERK, IRE1α, ATF6, and CHOP were all significantly reduced in the PBA+LPS group in comparison with those in the LPS group. Surprisingly, the LPS stimulation increased the level of phosphorylated AMPK, which was attenuated by inhibiting ER stress ($P < 0.01$, Fig-

ure 1b), indicating that AMPK may be involved in ER stress induced by LPS in bovine hepatocytes. The LPS treatment elevated the fluorescence intensity of GRP78 and CHOP, whereas 4-PBA pretreatment under LPS could alleviate this phenomenon (Figure 1d).

Inhibiting ER Stress Reduces LPS-Induced Autophagy and Mitochondrial Damage in Bovine Hepatocytes

Compared with that in the control group, LPS treatment upregulated the mRNA relative expression of autophagy-related genes, including unc-51-like kinase 1 (*ULK1*), BCL-2 interacting protein 1 (*BECLIN1),* autophagy-related gene 14 (*ATG14*), autophagy-related gene 12 (*ATG12*), autophagy-related gene 10 (*ATG10*), autophagy-related gene 5 (*ATG5*), autophagy-related gene 16-like 1 (*ATG16L1*), microtubule-associated protein 1 light chain 3 α (*MAP1LC3A*), and autophagy-related gene 4A (*ATG4A*; $P < 0.05$), but downregulated that of sequestosome 1 (*SQSTM1*, $P < 0.01$; Figure 2a). The 4-PBA pretreatment under LPS could reverse this trend (Figure 2a). In comparison with that in the LPS group, inhibiting ER stress reduced the protein abundance of BECLIN1, ATG5, LC3-II, lysosomal-associated membrane protein 2A (LAMP2A), syntaxin 17 (STX17), synaptosomal-associated protein 29 (SNAP29), and vesicle-associated membrane protein 8 (VAMP8; $P < 0.05$), whereas increased that of SQSTM1 ($P < 0.05$) in LPS-treated hepatocytes (Figure 2b). The fluorescence intensity of SQSTM1 was stronger in the PBA+LPS group, whereas that of LC3 in this group was weaker than that in the LPS group (Figure 2c). Furthermore, the LPS treatment increased the formation of lysosomes which could be reduced by 4-PBA pretreatment (Figure 2f). As shown in Figure 2d, pretreatment with 4-PBA under the LPS treatment downregulated the mRNA relative expression of apoptosis-related genes, including BCL2-associated X protein (*BAX*) and *CASPASE3* ($P < 0.01$), and upregulated that of B cell leukemia 2 (*BCL*-2; $P < 0.01$) in bovine hepatocytes. The western blot results showed that in comparison with that in the control group, the protein level of BAX, cleaved CASPASE3, and CASPASE3 ($P < 0.05$) was significantly increased, whereas that of BCL-2 was significantly decreased ($P < 0.05$) in the LPS-treated group, which could be mitigated by 4-PBA pretreatment (Figure 2e). Moreover, the 4-PBA pretreatment under the LPS treatment increased the mitochondrial membrane potential (**MMP**) reflected by the reduced green fluorescence and the increased red fluorescence intensities (Figure 2g). The LPS treatment inhibited mitofusin 2 (*MFN2*) expression and promoted the expression of *PARKIN* and *PINK1*, indicating mitochondrial dynamic disruption and mitophagy occurrence, whereas 4-PBA pretreatment under the LPS treatment could alleviate these phenomena (Figure 2h). The Mito-tracker staining results showed that treating hepatocytes with LPS switched the mitochondria appearance from a normal reticular shape to a short rod shape, which was improved by inhibiting ER stress with 4-PBA (Figure 2i).

Inhibition of AMPK Mitigates LPS-Induced ER Stress and Autophagy in Bovine Hepatocytes

The AMPK signaling pathway was reported to be the key regulator of ER stress, autophagy and mitochondrial function (Fan et al., 2020; Hu et al., 2021), and we speculated that AMPK might be responsible for the increased level of ER stress, autophagy, and mitochondrial dynamic. The inhibitor of AMPK, CC, was used to study the role of AMPK in mediating cellular homeostasis. The cell viability was not influenced by treatment with 0.1 to 1 μM CC for 9 h but was significantly decreased by the treatment with 2 and 5 μM CC ($P < 0.01$, Figure 3a). As shown in Figure 3b, compared with that in the LPS group, the protein expression of GRP78, PERK, IRE1α, ATF6 and CHOP was significantly diminished in the CC+LPS group ($P < 0.05$, Figure 3b). Immunofluorescence analysis showed that the GRP78 and CHOP proteins exhibited weaker staining in the CC+LPS group than shown in the LPS group (Figure 3c). Inhibiting AMPK reduced the protein expression of ULK1, ATG14, BECLIN1, LC3, SNAP29, and VAMP8 ($P < 0.05$), and enhanced the protein level of mechanistic target of rapamycin kinase (**mTOR**) and SQSTM1 ($P < 0.05$) in the LPS-treated bovine hepatocytes (Figure 3d). As showed in Figure 3e, the immunofluorescence intensity of SQSTM1 was increased in comparison with that in the LPS group while that of LC3 was decreased during CC pretreatment under LPS treatment. Lysosome formation increased after the LPS treatment, which was alleviated by the CC pretreatment (Figure 3f). Inhibition of AMPK restores mitochondrial homeostasis by enhancing mitochondrial fusion and reducing apoptosis in the LPS-treated hepatocytes

The western blot results showed that the expression of phosphoglycerate mutase family member 5 (PGAM5), dynamin-related protein 1 (DRP1), and PARKIN proteins was lower ($P < 0.01$) in the CC+LPS group than that in the LPS group, whereas that of the MFN2 protein was higher ($P < 0.05$; Figure 4a). The LPS stimulation reduced the immunofluorescence intensity of MFN2 protein and promoted that of PARKIN protein, whereas inhibiting AMPK with CC pretreatment under LPS treatment could reverse this trend (Figure

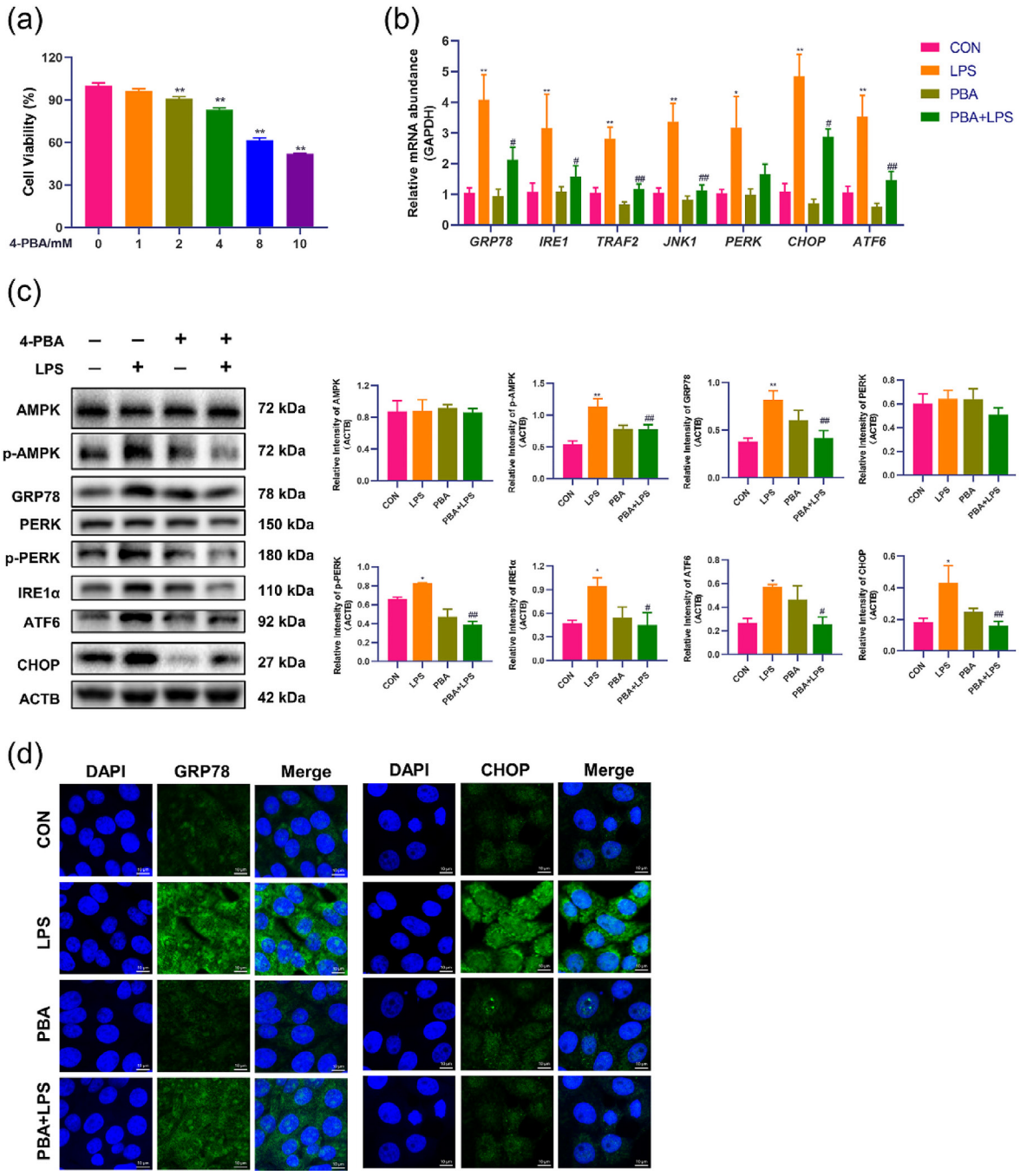


Figure 1. Effects of 4-phenylbutyric acid (4-PBA) on endoplasmic reticulum (ER) stress and adenosine 5′-monophosphate-activated protein kinase (AMPK) signaling pathway in LPS-treated bovine hepatocytes. (a) The percentage of cell viability during treatment of hepatocytes with different doses of 4-PBA. The cells were treated with different concentrations of 4-PBA for 12 h, and cell viability was tested with a CCK-8 kit. (b) The relative mRNA expression of ER stress-related genes in bovine hepatocytes. The cells were pretreated with 4-PBA (1 m*M*) for 12 h followed by a 12 μg/mL LPS treatment for another 12 h. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The PBA group was treated with 1 m*M* 4-PBA for 12 h. The PBA+LPS group was pretreated with 1 m*M* 4-PBA for 12 h followed by a 12 μg/mL LPS treatment for another 12 h. (c) The abundance of ER stress-related proteins in bovine hepatocytes. The cells were pretreated with 4-PBA (1 m*M*) for 12 h followed by the 12 μg/mL LPS treatment for another 12 h. Left panel represents the expression of target proteins analyzed by western blot, and right panel represents the relative intensity of target proteins. (d) Glucose-regulated protein (GRP78) and C/EBP homologous protein (CHOP) abundance visualized by fluoresceine isothiocyanate (FITC) fluorescence (green), and nuclei visualized by 4′,6-diamidino-2-phenylindole (DAPI; blue) through immunofluorescence staining were measured to study the effect of 4-PBA (1 m*M*) on the ER stress state of bovine hepatocytes. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. The data were analyzed using one-way ANOVA with Dunnett's post hoc test.

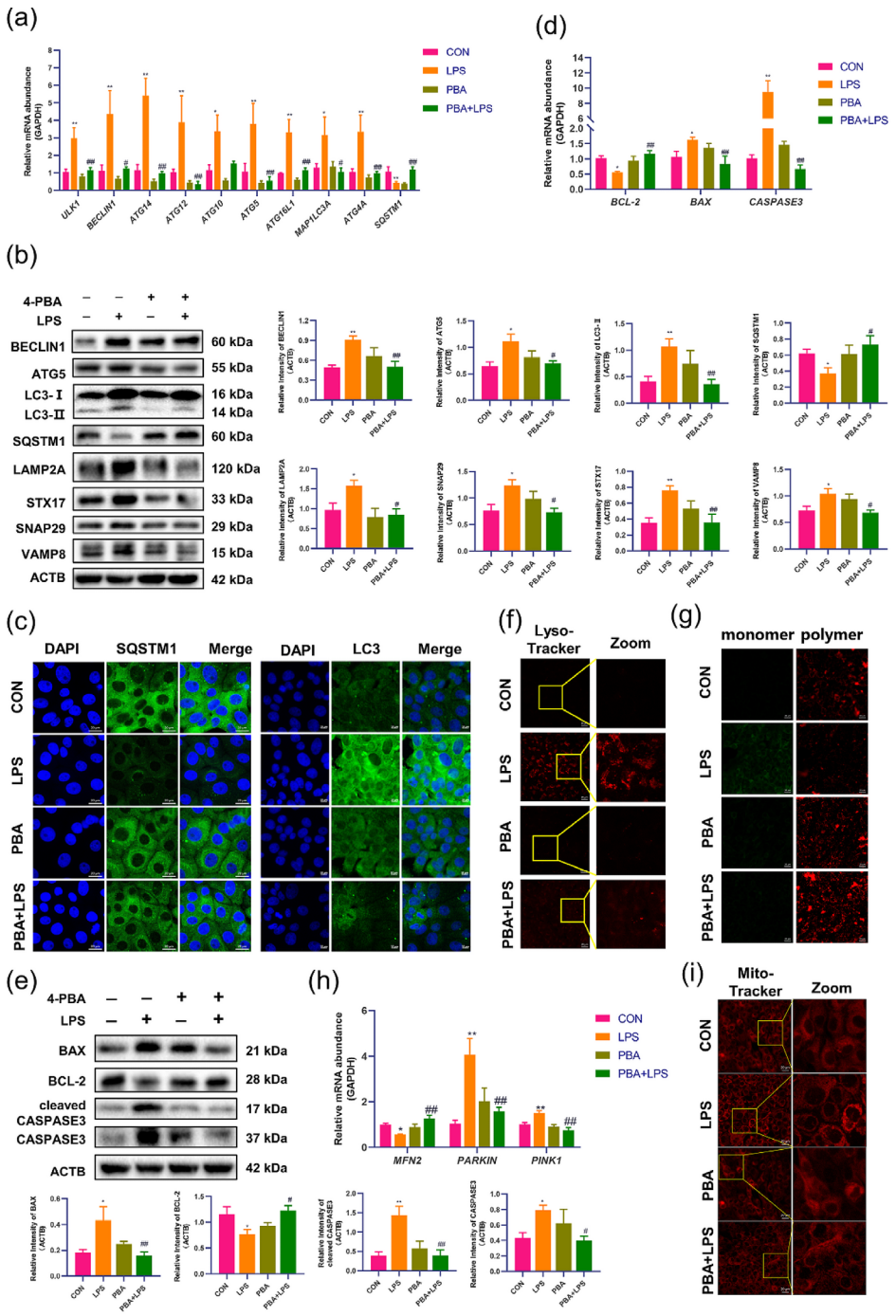


Figure 2. Inhibiting endoplasmic reticulum (ER) stress reduces autophagy and mitochondrial damage in the LPS-treated bovine hepatocytes. The cells were pretreated with 4-phenylbutyric acid (4-PBA; 1 m*M*) for 12 h followed by the 12 μg/mL LPS treatment for another 12 h. (a) The relative mRNA expression of autophagy-related genes in bovine hepatocytes. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The PBA group was treated with 1 m*M* 4-PBA for 12 h. The PBA+LPS group was pretreated with 1 mM 4-PBA for 12 h followed by a 12 μg/mL LPS treatment for another 12 h. (b) The abundance of autophagy-related proteins in bovine hepatocytes. (c) The protein expression of SQSTM1 and LC3 determined by immunofluorescence staining. The immunofluorescence analysis was performed with anti-SQSTM1 antibody (green) or anti-LC3 (green) antibody and 4′,6-diamidino-2-phenylindole (DAPI; blue; nuclei). (d) The relative mRNA expression of apoptosis-related genes. (e) The abundance of apoptosis-related proteins. (f) Lysosomes were stained with Lyso-tracker in bovine hepatocytes. (g) JC-1 staining was performed to analyze the mitochondrial membrane potential (MMP) in bovine hepatocytes. (h) The relative mRNA expression of mitochondrial dynamic-related genes. (i) Mitochondria were stained with Mito-tracker in bovine hepatocytes. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed using one-way ANOVA with Dunnett's post hoc test.

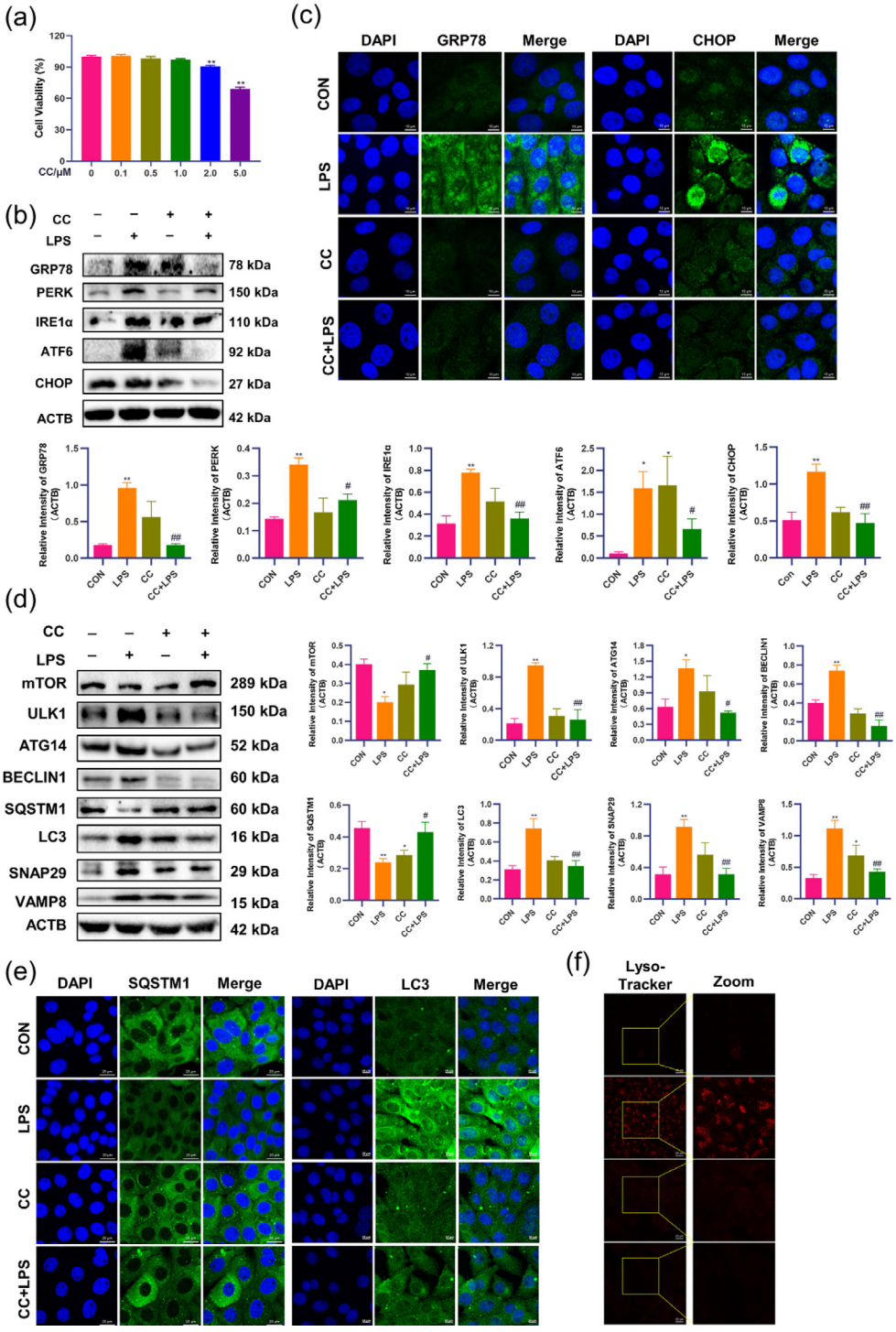


Figure 3. Inhibiting adenosine monophosphate-activated protein kinase (AMPK) alleviates endoplasmic reticulum (ER) stress and autophagy in LPS-treated bovine hepatocytes. Cells were pretreated with compound C (CC, 1 μM) for 9 h followed by 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during treatment of hepatocytes with different concentrations of CC for 9 h. (b) The abundance of ER stress-related proteins in bovine hepatocytes. (c) Immunofluorescence analysis of GRP78 and CHOP protein abundance. Green: GRP78 and CHOP; blue: nuclei. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The CC group was treated with 1 μM CC for 9 h. The CC+LPS group was pretreated with 1 μM CC for 9 h followed by a 12 μg/mL LPS treatment for another 12 h. (d) The abundance of autophagy-related proteins in bovine hepatocytes. (e) Immunofluorescence analysis of SQSTM1 and LC3 proteins. Green: SQSTM1 and LC3; blue: nuclei. (f) Lysosomes were stained with a Lyso-tracker in bovine hepatocytes. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed with one-way ANOVA with Dunnett's post hoc test.

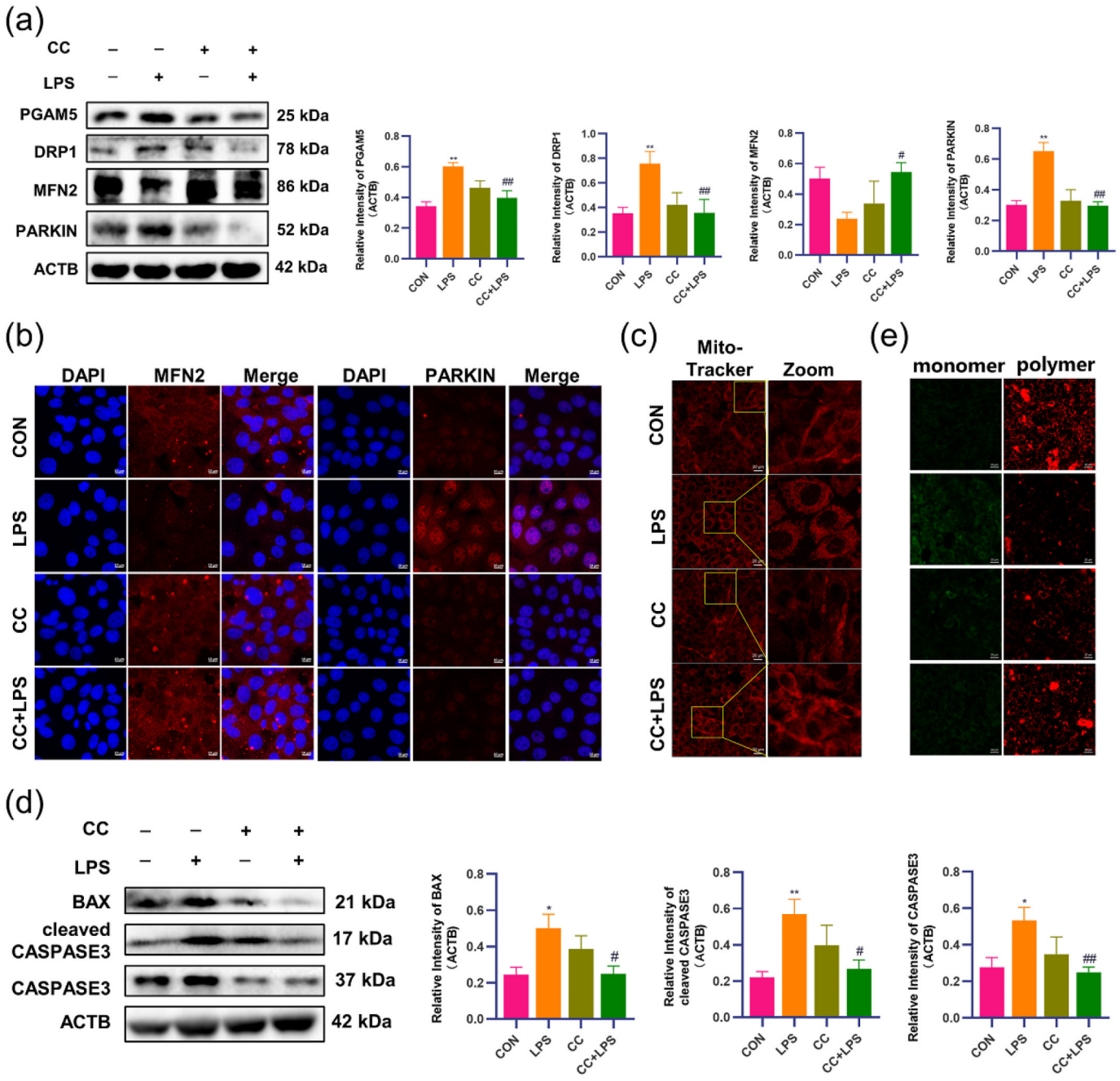


Figure 4. Inhibiting adenosine monophosphate-activated protein kinase (AMPK) mitigates mitochondrial damage in the LPS-treated bovine hepatocytes. The cells were pretreated with compound C (CC, 1 μM) for 9 h followed by the 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during the treatment of hepatocytes with different concentrations of CC for 9 h. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The CC group was treated with 1 μM CC for 9 h. The CC+LPS group was pretreated with 1 μM CC for 9 h followed by a 12 μg/mL LPS treatment for another 12 h. (b) The abundance of endoplasmic reticulum (ER) stress-related proteins in bovine hepatocytes. (c) Immunofluorescence analysis of GRP78 and CHOP proteins. Green: GRP78 and CHOP; blue: nuclei. (d) The abundance of autophagy-related proteins in bovine hepatocytes. (e) Immunofluorescence analysis of SQSTM1 and LC3 proteins. Green: SQSTM1 and LC3; blue: nuclei. (f) Lysosomes were stained with Lyso-tracker in bovine hepatocytes. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. The data were analyzed with one-way ANOVA with Dunnett's post hoc test.

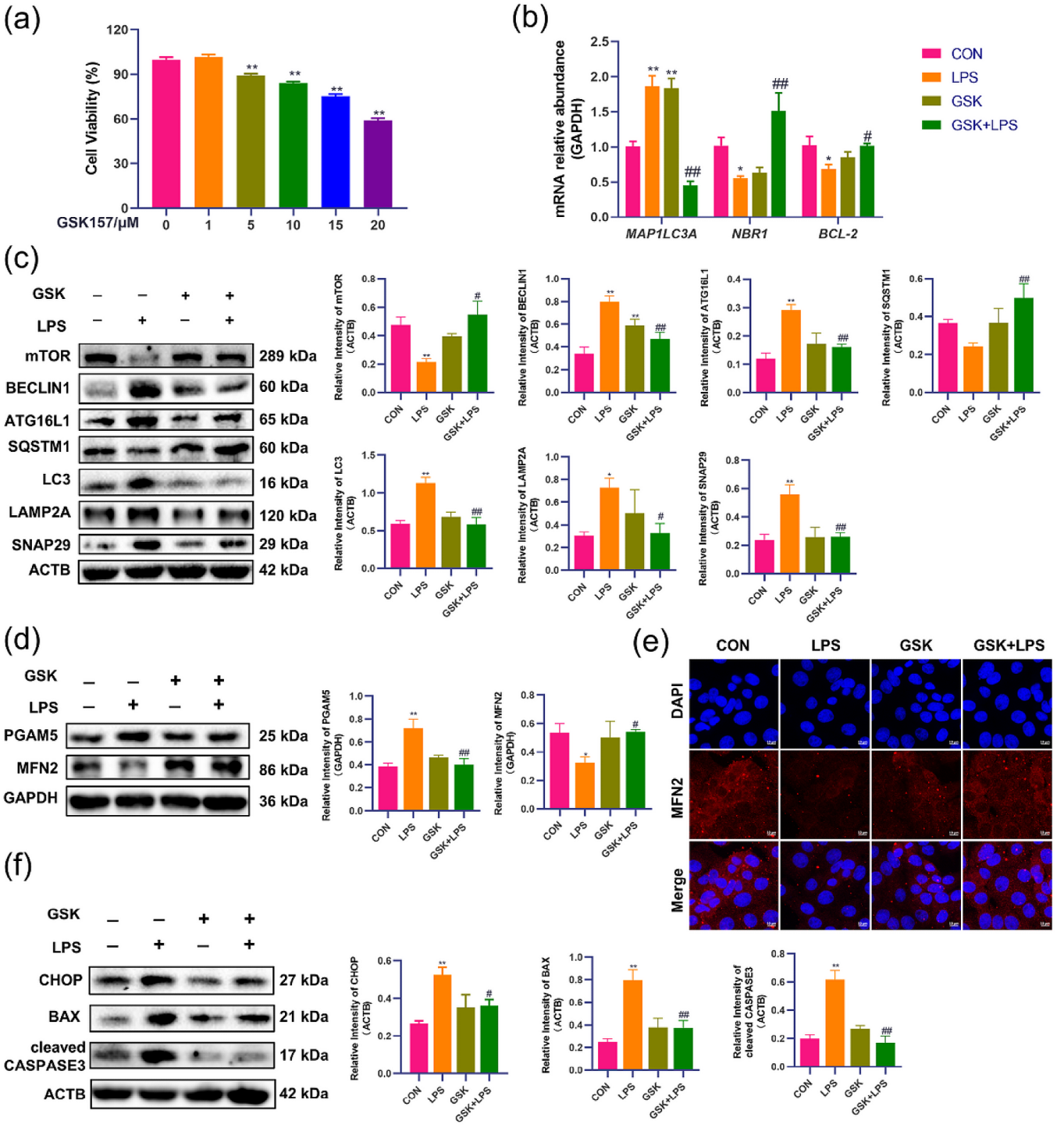


Figure 5. Inhibiting protein kinase-like endoplasmic reticulum kinase (PERK) relieves the LPS-induced autophagy and mitochondrial damage in bovine hepatocytes. The cells were pretreated with a PERK inhibitor, GSK157 (GSK, 1 μM), for 6 h followed by the 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during the treatment of hepatocytes with different concentrations of GSK for 6 h. (b) The mRNA relative abundance of autophagy and apoptosis-related genes in bovine hepatocytes. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The GSK group was treated with 1 μM GSK for 6 h. The GSK+LPS group was pretreated with 1 μM GSK for 6 h followed by a 12 μg/mL LPS treatment for another 12 h. (c) The abundance of autophagy-related proteins. (d) The abundance of mitochondrial dynamic-related proteins. (e) Immunofluorescence analysis of MFN2 protein. Red: MFN2; blue: nuclei. (f) The abundance of apoptosis-related proteins. The data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed with one-way ANOVA with Dunnett's post hoc test.

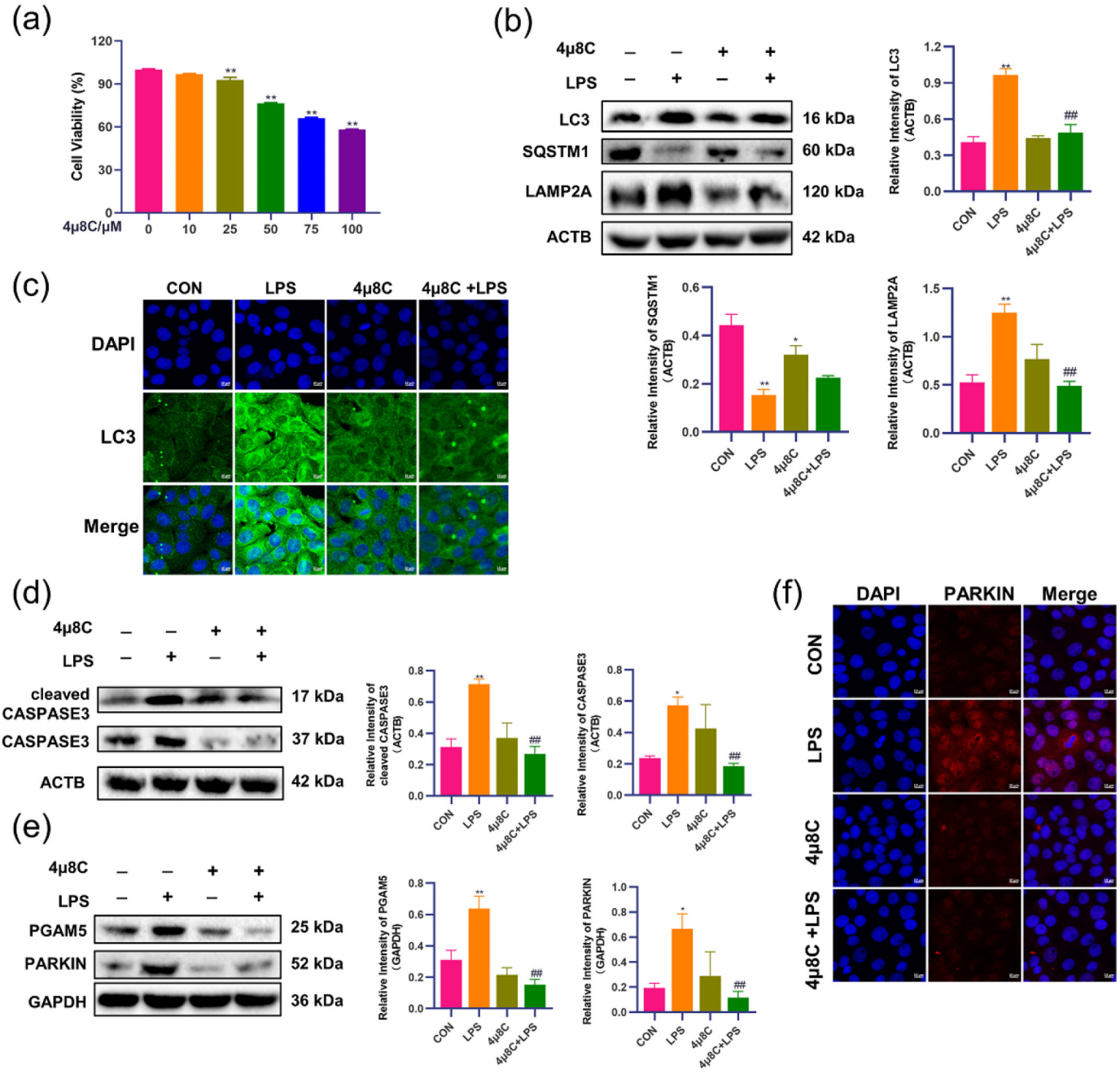


Figure 6. Inhibition of IRE1α reduces autophagy and mitochondrial damage in LPS-treated bovine hepatocytes. The cells were pretreated with an IRE1 inhibitor, 4μ8C (10 μ*M*), for 12 h followed by the 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during the treatment of hepatocytes with different concentrations of 4μ8C for 12 h. (b) The abundance of autophagy-related proteins in bovine hepatocytes. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The 4μ8C group was treated with 10 μ*M* 4μ8C for 12 h. The 4μ8C+LPS group was pretreated with 10 μ*M* 4μ8C for 12 h followed by a 12 μg/mL LPS treatment for another 12 h. (c) The LC3 protein abundance analyzed by immunofluorescence analysis. Green: LC3; blue: nuclei. (d) The abundance of apoptosis-related proteins. (e) The abundance of mitochondrial dynamic-related proteins. (f) Immunofluorescence analysis of PARKIN protein. Red: PARKIN; blue: nuclei. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed with one-way ANOVA with Dunnett's post hoc test.

4b). The normal reticular shape of mitochondria was restored in the CC+LPS group but remained as short rods in the LPS group (Figure 4c). Furthermore, inhibiting AMPK downregulated the expression of BAX, cleaved CASPASE3 and CASPASE3 ($P < 0.05$, Figure 4d), and elevated MMP (Figure 4e) to mitigate LPS-induced apoptosis in bovine hepatocytes.

Inhibiting PERK Alleviates the LPS-Induced Autophagy and Mitochondrial Damage in Bovine Hepatocytes

A previous study has showed that PERK plays a vital role in autophagy and mitochondrial fusion via interaction with MFN2 (Verfaillie et al., 2012). Herein, we used a PERK inhibitor to examine whether PERK is involved in the LPS-caused mitochondrial dysfunction and autophagy. The results showed that the cell viability was not influenced by the 1 μM GSK treatment for 12 h but was significantly decreased at the doses of 5, 10, 15, and 20 μM GSK treatment ($P < 0.01$, Figure 5a). Hence, we chose the 1 μM dose to explore the role of PERK in regulating autophagy and mitochondrial damage. As shown in Figure 5b, inhibiting PERK could reduce the mRNA expression of *MAP1LC3A* ($P < 0.01$), and promote that of neighbor of BRCA1 gene 1 protein (*NBR1*) and *BCL*-2 ($P < 0.05$) in the LPS-treated hepatocytes. Compared with that in the LPS group, the protein level of BECLIN1, ATG16L1, LC3, LAMP2A, and SNAP29 was significantly decreased ($P < 0.05$), whereas that of mTOR and SQSTM1 was increased ($P < 0.05$) in the GSK+LPS group. The protein abundance of PGAM5 was downregulated ($P < 0.01$) and that of MFN2 was upregulated ($P < 0.05$) in the GSK+LPS group in comparison with the LPS group. The immunofluorescence analysis showed that the MFN2 protein exhibited a stronger staining in the GSK+LPS group compared with that in the LPS group (Figure 5e). Inhibiting PERK also diminished the expression levels of CHOP, BAX, and CASPASE3 ($P < 0.05$) to reduce the LPS-induced apoptosis in bovine hepatocytes (Figure 5f).

Inhibiting IRE1α Reduces Autophagy and Mitochondrial Damage in LPS-Treated Bovine Hepatocytes

As shown in Figure 6a, a 4μ8C treatment for 12 h significantly reduced the cell viability for the 25, 50, 75, and 100 μM doses ($P < 0.01$); hence, the 10 μM dose was chosen for further analysis. Inhibiting IRE1α reduced the expression of LC3 and LAMP2A proteins ($P < 0.01$), and promoted that of SQSTM1 in the LPS-treated hepatocytes (Figure 6b). The immunofluorescence intensity of LC3 was increased by an LPS administration but was reduced by the 4μ8C pretreatment in hepatocytes (Figure 6c). Compared with that in the LPS group, the expression of CASPASE3 and cleaved CASPASE3 was significantly downregulated in the 4μ8C+LPS group ($P < 0.01$, Figure 6d). The protein levels of PGAM5 and PARKIN was significantly upregulated by the LPS treatment ($P < 0.05$), which could be alleviated by the 4μ8C pretreatment in the LPS-treated hepatocytes (Figure 6e). The immunofluorescence staining results showed that the immunofluorescence intensity of PARKIN was lower in the 4μ8C+LPS group than that in the LPS group (Figure 6f).

Inhibiting JNK Ameliorates LPS-Induced Mitophagy and Mitochondrial Damage in Bovine Hepatocytes

Herein, the cell viability during treatment with 10 μM SP125 for 12 h was not influenced by SP125 but it was significantly reduced at the 25, 50, 75, and 100 μM doses ($P < 0.01$, Figure 7a). The mRNA level of *CHOP*, *ATG5*, and *MAP1LC3A* was significantly downregulated ($P < 0.01$) and that of *NBR1* was upregulated ($P < 0.01$) in the SP125+LPS group in comparison with that in the LPS group (Figure 7b). Compared with that in the LPS group, the protein level of BECLIN1, LC3, and LAMP2A in the SP125+LPS group was significantly decreased ($P < 0.01$), whereas that of mTOR was significantly increased ($P < 0.01$) in the SP125+LPS group (Figure 7c). As shown in Figure 7d, the protein level of BAX and CASPASE3 was significantly lower in the SP125+LPS group than in the LPS group ($P < 0.01$, Figure 7d). Furthermore, inhibiting JNK downregulated the expression of mitophagy-related proteins (PINK1 and PARKIN) in LPS-treated hepatocytes ($P < 0.05$, Figure 7e). The immunofluorescence intensity of PARKIN was decreased in the SP125+LPS group compared with that in the LPS group (Figure 7f).

Blockage of Autophagy Alleviates LPS-Induced Mitochondrial Damage in Bovine Hepatocytes

To explore the effect of autophagy on mitochondrial damage in LPS-treated bovine hepatocytes, the autophagy inhibitor CQ was used in current study. Cell viability was significantly decreased with 10 to 100 μM CQ treatment for 6 h ($P < 0.05$, Figure 8a). Pretreating cells with 1 μM CQ for 6 h significantly downregulated the protein abundance of ULK1 and LAMP2A ($P < 0.05$) and upregulated that of mTOR and SQSTM1 ($P < 0.01$) in LPS-treated hepatocytes (Figure 8b). The SQSTM1 protein exhibited stronger staining in the CQ+LPS group than that in the LPS group

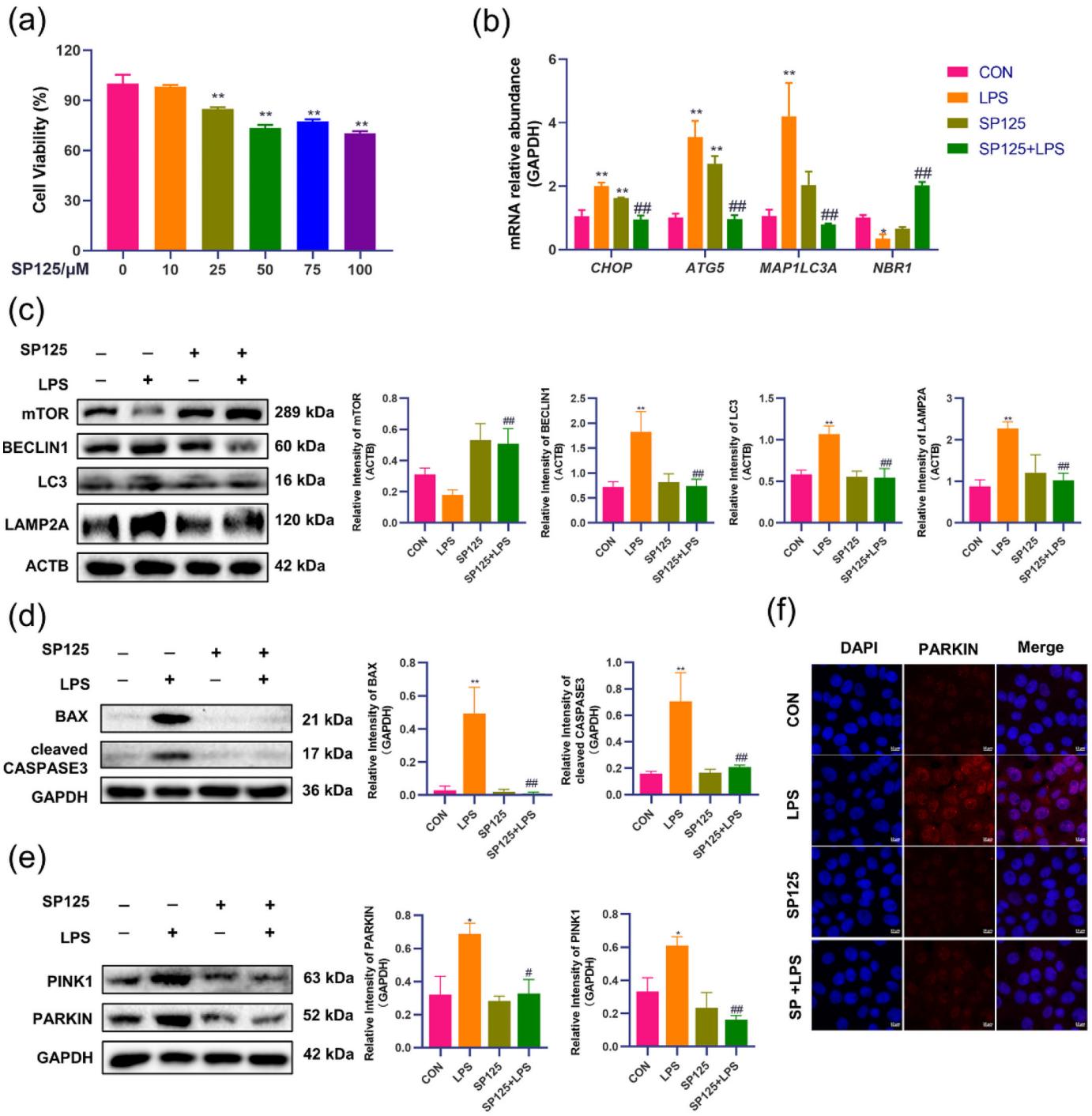


Figure 7. Inhibition of JNK alleviates the autophagy and mitochondrial damage caused by LPS in bovine hepatocytes. The cells were pretreated with a JNK inhibitor, SP125 (10 μ*M*), for 12 h followed by 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during treatment of hepatocytes with different concentrations of SP125 for 12 h. (b) The mRNA expression of apoptosis and autophagy. (c) The abundance of autophagy-related proteins. (d) The abundance of apoptosis-related proteins. (e) The abundance of mitophagy-related proteins. (f) Immunofluorescence analysis of PARKIN protein. Red: PARKIN; blue: nuclei. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The SP125 group was treated with 10 μ*M* SP125 for 12 h. The SP125+LPS group was pretreated with 10 μ*M* SP125 for 12 h followed by a 12 μg/mL LPS treatment for another 12 h. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed with one-way ANOVA with Dunnett's post hoc test.

(Figure 8d). Moreover, the CQ pretreatment inhibited the formation of lysosomes in the LPS-treated hepatocytes (Figure 8e). The western blot results showed that the LPS treatment increased the protein expression of BAX, cleaved CASPASE3, and CASPASE3 ($P < 0.05$) and decreased that of BCL-2 ($P < 0.05$) in bovine hepatocytes, whereas autophagy blockage with CQ pretreatment could mitigate these changes in LPS-treated hepatocytes (Figure 8c). Moreover, the MMP was increased during the CQ pretreatment under LPS administration (Figure 8f). As shown in Figure 8f, the CQ pretreatment switched mitochondria from an abnormal short rod shape to a normal reticular shape in LPS-treated cells.

DISCUSSION

Mitochondria rich in hepatocytes are important for maintaining hepatic homeostasis, whereas mitochondrial damage contributes to liver dysfunction (Teckman et al., 2004; McLaughlin et al., 2020). In response to cellular stress, ER stress is initiated to maintain homeostasis, and ER stress is associated with mitochondrial dysfunction via MAM to trigger prosurvival/prodeath pathways (Giorgi et al., 2015). It has been reported that AMPK is the energy sensor and play a vital role in maintaining cellular homeostasis via mediating mitochondrial biogenetics, ER stress, and autophagy (Sun et al., 2019; Fan et al., 2020; Hu et al., 2021). Study shows that valdecoxib treatment increases the expression of AMPK phosphorylation and ameliorates ER stress, whereas siRNA-mediated suppression of AMPK abolishes the effects of valdecoxib on the ER stress by upregulating the expression of GRP78, p-eIF2α, and CHOP in palmitate-challenged human primary hepatocytes (Park et al., 2022). An AMPK inhibitor pretreatment abrogated the increased expression of the ER stress- and autophagy-associated signals in the chicoric acid-treated gastric cancer cells, indicating that AMPK activation is essential for initiating ER stress and autophagy (Sun et al., 2019). Herein, we demonstrated that inhibiting AMPK alleviated the LPS-caused ER stress by diminishing the expression level of the ER stress-related genes. Under energy stress conditions, AMPK interacts with MFN2 to promote the MAM formation and autophagy for autophagic removal of mitochondria in metformin- or rotenone-treated Hela and U2OS cells (Hu et al., 2021). Recent publications report that both PERK and IRE1α localize to MAM to sustain the ER-mitochondria communication during various cellular stress (Verfaillie et al., 2012; Mori et al., 2013; Carreras-Sureda et al., 2019). Hence, AMPK may control MAM function by regulating the expression of PERK, IRE1α, and MFN2 to be involved in the LPS-caused disruption of mitochondrial homeostasis, as shown by the downregulated expression of PERK and IRE1α, and the upregulated expression of MFN2 under AMPK inhibitor pretreatment in the LPS-treated hepatocytes. Inhibiting AMPK can alleviate inorganic arsenic-induced autophagy and apoptosis activation by downregulating the expression of cleaved CASPASE3 and LC3-II, and inhibiting the formation of acidic vesicular organelles (Fu et al., 2021). Another research demonstrates that AMPK is involved in citrinin-induced apoptosis and inhibiting AMPK can significantly decrease the apoptosis level in L02 cells (Wu et al., 2022). Consistently, we found that inhibiting AMPK reduced the levels of autophagy and apoptosis as well as maintained the balance between mitochondrial fusion and fission. Hence, AMPK activation plays a vital role in the LPS-induced ER stress and mitochondrial homeostasis. Growing evidence has shown that autophagy is initiated by the activated AMPK and the inactivated mTOR, which can enhance the ULK1 activity in response to cellular stimulation (Kim et al., 2011; Wu et al., 2021). Herein, inhibiting AMPK inhibited the LPS-caused decreased expression of mTOR and increased expression of ULK1, indicating that AMPK may be involved in autophagy by inactivating mTOR and promoting the ULK1 expression in the LPS-treated hepatocytes.

The publication demonstrates that cantharidin treatment induces ER stress, autophagy and apoptosis, and the ER stress inhibitor 4-PBA can downregulate ER stress-, autophagy- and apoptosis-related genes expression in HK-2 cells, indicating that ER stress is responsible for the activation of autophagy and apoptosis (He et al., 2022). Blocking ER stress with its antagonist ameliorates the 3-MCPD-induced ER stress by inhibiting the PERK and IRE1 branches as well as the mitochondrial apoptosis by regulating mitochondrial fission and fusion in HEK293 cells (Zhong et al., 2021). Previous study has demonstrated that tunicamycin-induced ER stress causes the loss of MMP, mitochondrial cytochrome c release in HT22 cells (Song et al., 2016). In prostate cancer cells, inhibiting ER stress with 4-PBA abolishes the gambogenic acid-induced apoptosis and autophagy via the JNK pathway (Wu et al., 2023). Herein, inhibiting ER stress with 4-PBA alleviated autophagy and mitochondrial dysfunction in the LPS-treated bovine hepatocytes, indicating that ER stress contributed to the LPS-induced autophagy and apoptosis.

The induction of the IRE1α-JNK and PERK pathways is the dominated ways to enhance autophagy and apoptosis (Song et al., 2017). Additionally, both PERK and IRE1α localize to MAM to sustain the ER-mitochondria communication in response to diverse cellular

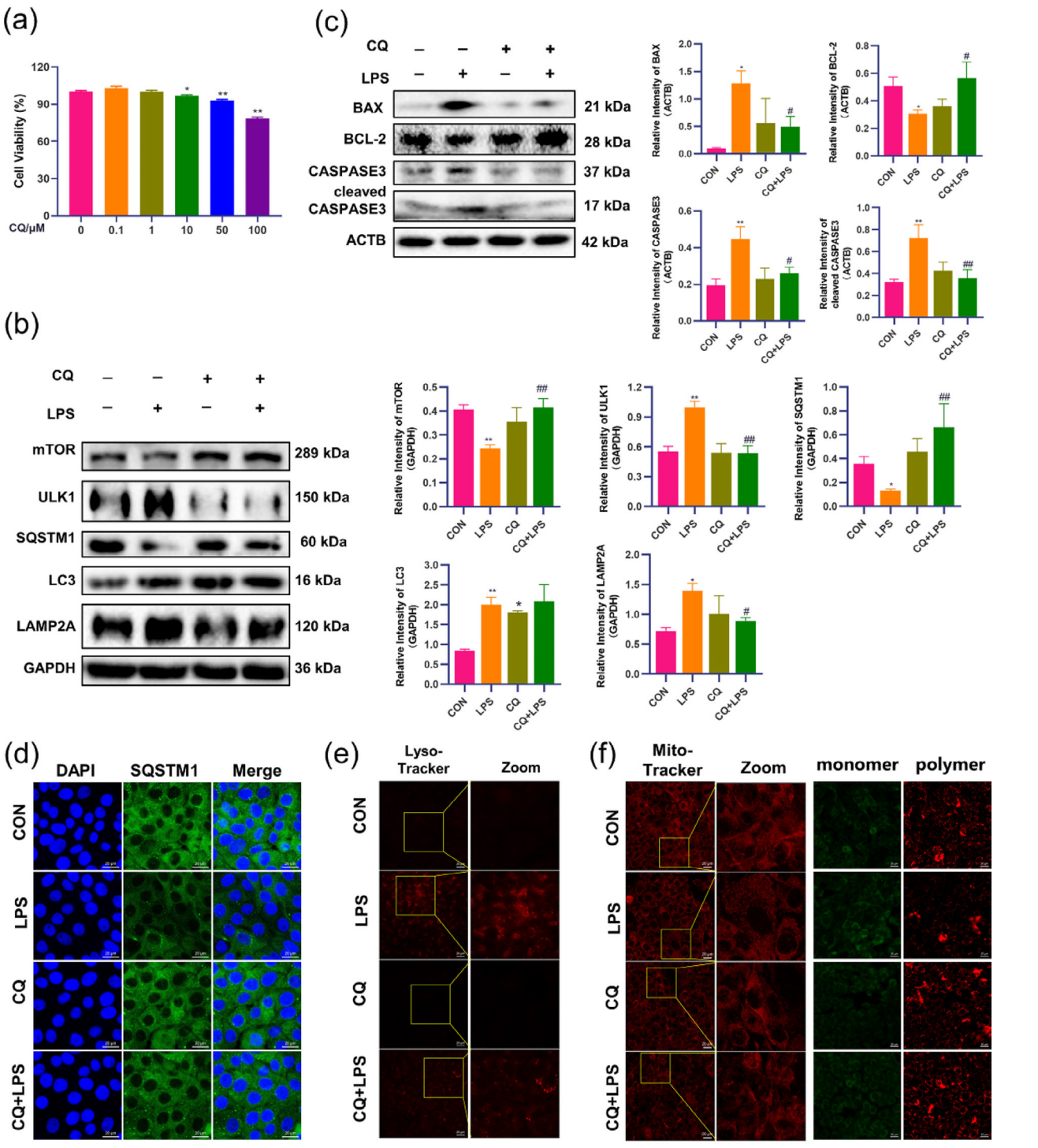


Figure 8. Inhibition of autophagy relieves the LPS-induced mitochondrial damage in bovine hepatocytes. The cells were pretreated with an autophagy inhibitor, chloroquine (CQ, 1 μM), for 6 h followed by the 12 μg/mL LPS treatment for another 12 h. (a) The percentage of cell viability during treatment of hepatocytes with different concentrations of CQ for 6 h. (b) The abundance of autophagy-related proteins. The CON group did not receive any treatment. The LPS group was treated with 12 μg/mL LPS for 12 h. The CQ group was treated with 1 μM CQ for 6 h. The CQ+LPS group was pretreated with 1 μM CQ for 6 h followed by a 12 μg/mL LPS treatment for another 12 h. (c) The abundance of apoptosis-related proteins. (d) Immunofluorescence analysis of SQSTM1 protein. Green: SQSTM1; blue: nuclei. (e) Lysosomes were stained with Lyso-tracker in bovine hepatocytes. (f) Mitochondria were stained with Mito-tracker and MMP was analyzed by JC-1 staining in bovine hepatocytes. Data are presented as the means ± SEM. Experiments are representative of 3 independent experiments (n = 3). $^{*}P < 0.05$ and $^{**}P < 0.01$ significantly different from control group; $\#P < 0.05$ and $\#\#P < 0.01$ significantly different from LPS. Data were analyzed with one-way ANOVA with Dunnett's post hoc test.

stress (Verfaillie et al., 2012; Mori et al., 2013; Carreras-Sureda et al., 2019). Initially, PERK was identified as a crucial member of the MAM, and it transmits apoptotic signals from the ER to mitochondria (Kwak et al., 2020). The downstream target of PERK is CHOP, a proapoptotic gene that suppresses the transcription of BCL-2 genes and stimulates the expression of BAX and BIM (Bravo-Sagua et al., 2013). In mouse embryo fibroblasts, PERK serves as a component of the MAM, and PERK deficiency mitigates the ROS-induced ER stress-mediated apoptosis by reducing the perturbations of the ER-mitochondria contact sites (Verfaillie et al., 2012). In high-concentration glucose-treated podocytes, MFN2 physically interacts with PERK. The inhibition of PERK phosphorylation restores mitochondrial functions and apoptosis by restoring the MMP, reducing ROS generation, and upregulating the ATP levels, but has no effect on the expression of MFN2 (Cao et al., 2021). Herein, inhibiting PERK decreased the expression of autophagy-related proteins and that of CHOP, BAX, and cleaved CASPASE 3 to reduce the levels of autophagy and apoptosis. Furthermore, the mitochondrial dynamic balance was restored by downregulating PGAM5 expression and upregulating the MFN2 expression in LPS-treated bovine hepatocytes. Another study shows that the PERK-ATF4-CHOP pathway is involved in triptolide-induced autophagy and apoptosis, and the PERK inhibitor GSK2656157 treatment downregulates the LC3 expression and upregulates SQSTM1 expression in the HepaRG cell line (Zhang et al., 2022). Moreover, PERK participates in the nonesterified fatty acids-induced autophagy in bovine hepatocytes, and inhibiting PERK diminishes the expression of ATG7, LC3-II, and BECLIN1, as well as the formation of autophagosomes (Huang et al., 2021). Thus, PERK might interact with MFN2 to regulate the MAM function, further modulating autophagy and apoptosis in bovine hepatocytes.

Recently, IRE1α was found to localize to MAM and directly impact the mitochondrial bioenergetics in mammalian cells (Carreras-Sureda et al., 2019). In melanoma cells, the activation of IRE1α is associated with mitochondrial fission and mitophagy by promoting the ubiquitination and degradation of MFN2 in

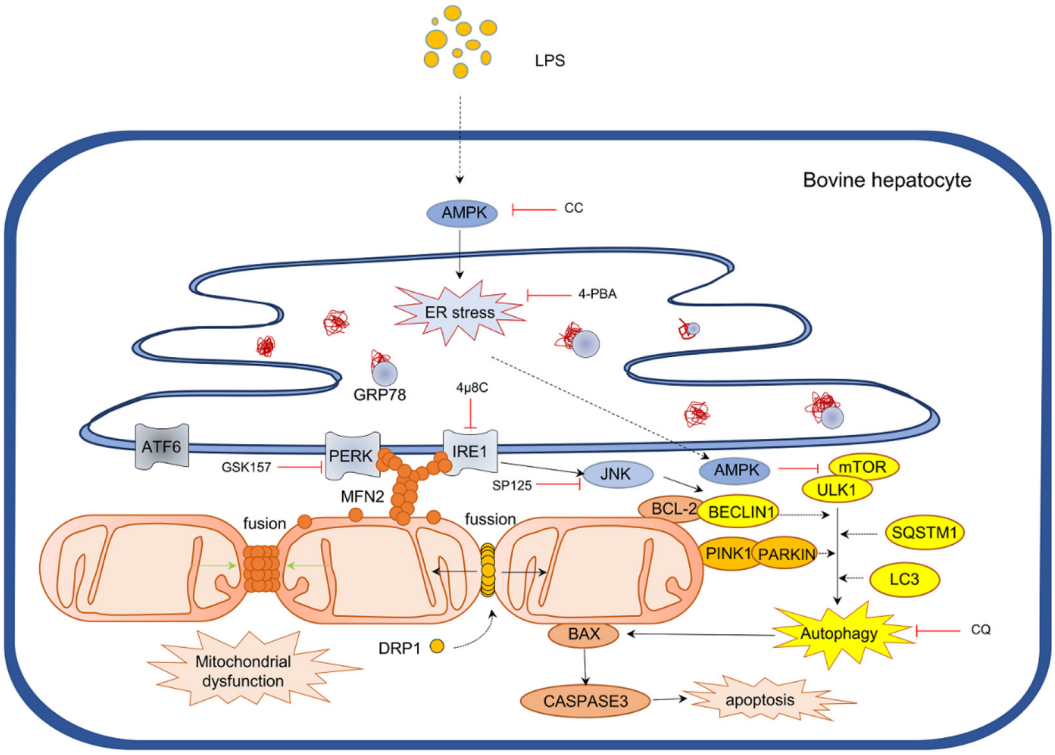


Figure 9. Schematic process of the AMPK-ER stress axis-mediated mitochondrial homeostasis in the LPS-treated bovine hepatocytes. ER = endoplasmic reticulum; AMPK = adenosine 5′-monophosphate-activated protein kinase; GRP78 = 78 kDa glucose-regulated protein; ATF6 = activating transcription factor 6; PERK = RNA-activated protein kinase (PKR)-like ER kinase; IRE1 = inositol-requiring enzyme 1; JNK = c-Jun N-terminal kinase; BCL-2 = B cell leukemia 2; PINK1 = PTEN induced putative kinase 1; BAX = BCL2-associated X protein; mTOR = mechanistic target of rapamycin kinase; ULK1 = unc-51-like kinase 1; SQSTM1 = sequestosome 1; LC3 = microtubule-associated protein 1 light chain 3; DRP1 = dynamin-related protein 1; MFN2 = mitofusin 2; 4-PBA = 4-phenylbutyric acid; CC = compound C; GSK157 = GSK2656157; SP125 = SP600125; CQ = chloroquine.

melanoma cells (Wang et al., 2021). The IRE1 deletion results in the accumulation of misfolding proteins in mitochondrial intermembrane space (Sarkar et al., 2022). Inhibition of the IRE1-JNK-CHOP pathway suppresses cell death and autophagy by downregulating the expression of cleaved CASPASE 12 and LC3B in kaempferol-induced gastric cancer cells (Kim et al., 2018). Moreover, IRE1α knockdown blocks autophagic flux, and inhibition of autophagy protects LX-2 cells from apoptotic death, indicating that the IRE1α pathway is essential for caffeine-induced autophagy and apoptosis in hepatic stellate cells (Li et al., 2017). In dioscin-treated diabetic rats, the expression of IRE1 and p-IRE1 is decreased with the increased levels of autophagy and mitophagy, and the enhanced ability of mitochondrial fusion in kidney (Zhong et al., 2022). We found that inhibiting IRE1α downregulated the expression of LC3 protein but upregulated the expression of SQSTM1 protein expression to block autophagic flux. Inhibiting IRE1α also reduced the abundance of cleaved CASPASE3, CASPASE3, PGAM5, and PARKIN to restore mitochondrial function. Therefore, IRE1α may be involved in the LPS-caused autophagy and mitochondrial damage by disrupting MAM function in bovine hepatocytes.

In the early phase of ER stress, JNK-mediated phosphorylation of BCL-2 results in the dissociation of the BECLIN1/BCL-2 and the activation of autophagy. Previous reports have demonstrated that the ER stress-induced IRE1/JNK pathway results in BCL-2/BECLIN1 inhibitory interactions leading to autophagy (Corazzari et al., 2015; He et al., 2021). We investigated the role of JNK in regulating autophagy and apoptosis by using a JNK inhibitor in LPS-treated hepatocytes. The results demonstrated that inhibiting JNK could restore mitochondrial homeostasis by decreasing the expression of apoptosis- and mitophagy-related proteins, and reduce the autophagy level by downregulating the expression of autophagy-related proteins, including BECLIN1 and LC3 in LPS-treated hepatocytes. Consistent with these results, inhibiting ER stress with 4-PBA abolished JNK activation, whereas the JNK inhibitor treatment could attenuate apoptosis and autophagy in the gambogenic acid-challenged prostate cancer cells (Wu et al., 2023). The application of JNK inhibitor SP600125 diminishes BECLIN1 expression and mitigates autophagy in phloretin-treated SH-SY5Y cells (Fan et al., 2022). The JNK activation and JNK mitochondrial localization are induced in tunicamycin-treated A549 cells and HUVEC or LPS-treated mice, and inhibiting JNK can alleviate apoptosis and autophagy to restore mitochondrial homeostasis, indicating that ER stress-JNK contributes to the LPS-caused mitochondrial dysfunction (Li et al., 2022). Moreover, JNK inhibition reduces the level of apoptosis and enhances MMP during *Trichomonas vaginalis*-induced ER stress in SiHa cells (Gao et al., 2021). Therefore, JNK may be involved in BECLIN1/BCL-2 dissociation to participate in LPS-induced mitochondrial dysfunction and autophagy in bovine hepatocytes.

The autophagy induced by low concentrations of crotonaldehyde enhances apoptosis, and autophagy blockage by bafilomycin A1 reduces the expression of cleaved CASPASE3 and cleaved CASPASE9, indicating that autophagy induces apoptosis via the mitochondria-mediated pathway in human bronchial epithelial cells (Wang et al., 2019). Another study shows that inhibiting autophagy with 3-MA can attenuate the palmitic acid- and ER stress-induced apoptosis by downregulating the apoptosis-related gene expression (*CASPASE 3* and *BAX*) in Saos-2 cells (Yang et al., 2018). Moreover, 3-MA can attenuate inorganic arsenic-induced apoptosis by reducing the activation of CASPASE3 activity in Neuro-2a cells (Fu et al., 2021). These studies support our results that inhibiting autophagy could reduce apoptosis to alleviate mitochondrial dysfunction in the LPS-treated bovine hepatocytes. Therefore, autophagy contributed to apoptosis in the LPS-treated bovine hepatocytes.

CONCLUSIONS

The AMPK-ER stress axis contributed to the LPS-induced mitochondrial dysfunction and autophagy, where the PERK/IRE1α-mediated MAM dynamic played a vital role in maintaining cellular homeostasis in bovine hepatocytes (Figure 9). Our study provides a theoretical basis for a novel approach to control the negative effects of LPS on the liver of dairy cows.

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